Unified Q_{μν} Framework & EFT Pipeline

This document compiles the core structures, decisions, and rules we solidified in the conversation. It is the base spec for the G-Theory / EFT quantum-corrections program.

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0. Meta: Epistemic Rules

All work obeys these rules:

- **No bullshit**: if something is not derived or computed, it is labeled **UNPROVEN**.
- **EFT first**: prefer standard effective field theory (EFT) and semiclassical gravity over numerology or speculative structures.
- **Explicit decomposition**: always separate
- well-defined geometric corrections,
- semiclassical field corrections,
- speculative add-ons.
- **Scaling sanity check**: before chasing observables, apply dimensional / curvature scaling. If suppression is ~$(M_{\rm Pl}/M)^4$ for astrophysical scales, it is observationally dead.
- **Speculation quarantine**: speculative constructs (primes, exotic entropy functionals, etc.) live in an explicit **Speculative Archive** and never share the same status as EFT pieces.

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1. Master EFT Action and Parameters

We fix the gravitational sector to a minimal, physically motivated quadratic gravity EFT:

$$\mathcal{L}_{\rm grav} = \frac{M_{\rm Pl}^2}{2}\Big[-R + \frac{1}{6m_0^2}R^2 - \frac{1}{2m_2^2}C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma}\Big]$$

- m_0: mass of the scalar (spin-0) mode from the R^2 term.
- m_2: mass of the massive spin-2 mode from the C^2 term.

Relations to traditional couplings (schematically):

- $\alpha \sim M_{\rm Pl}^2/(12m_0^2)$ for R^2,
- $\beta \sim -M_{\rm Pl}^2/(4m_2^2)$ for C^2.

Convention: work in Planck units where convenient: $M_{\rm Pl} = 1, G = 1/(8\pi), c = \hbar = 1$.

The total effective action is

$$S_{\rm eff} = \int d^4x\,\sqrt{-g}\,\Big[\mathcal{L}_{\rm grav} + \mathcal{L}_{\rm matter}\Big] + S_{\rm boundary} + S_{\rm torsion} + S_{\rm speculative}.$$

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2. Canonical Q_{μν} Template

All gravitational equations are written as:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} + Q_{\mu\nu} = 8\pi G\, T_{\mu\nu}^{\text{classical}}.$$

We **always** decompose:

$$Q_{\mu\nu} = Q_{\mu\nu}^{\text{geom}} + Q_{\mu\nu}^{\text{field}} + Q_{\mu\nu}^{\text{spec}}.$$

2.1 Geometric EFT correction Q^{geom}_{μν}

Defined by variation of higher-curvature terms only:

$$Q_{\mu\nu}^{\text{geom}} \equiv -\frac{2}{\sqrt{-g}}\frac{\delta}{\delta g^{\mu\nu}} \int d^4x\sqrt{-g}\Big[\frac{M_{\text{Pl}}^2}{12m_0^2}R^2 - \frac{M_{\text{Pl}}^2}{4m_2^2}C_{\alpha\beta\gamma\delta}C^{\alpha\beta\gamma\delta}\Big].$$

- This is the **EFT geometric correction** sector.
- In conformally flat backgrounds (like FLRW), the C^2 term vanishes, so only R^2 contributes.

2.2 Field / semiclassical correction Q^{field}_{μν}

Includes expectation values and anomaly-type terms:

$$Q_{\mu\nu}^{\text{field}} \equiv 8\pi G\Big(\langle \hat{T}_{\mu\nu}\rangle_{\text{ren}} + T_{\mu\nu}^{(\phi)} + T_{\mu\nu}^{\text{anomaly}} + \dots\Big).$$

Examples: - Renormalized stress-energy of quantum fields on curved backgrounds. - Effective scalaron stress tensor in Einstein frame (if desired). - Conformal anomaly stress tensor for conformal fields.

2.3 Speculative sector Q^{spec}_{μν}

Reserve this for constructs not directly derived from EFT/QFT:

$$Q_{\mu\nu}^{\text{spec}} \equiv [\text{Speculative terms: prime-weighted curvature, ad-hoc entropy functionals, etc.}]$$

Rules: - Must be explicitly identified as **UNPROVEN**. - Must not be mixed into predictions without clear labeling. - Many such constructs (e.g., PWG for BHs) are now known to be **phenomenologically inert**.

2.4 Epistemic status

- $Q_{\mu\nu}^{\text{geom}}$: **Established EFT** (up to choice of m_0, m_2).
- $Q_{\mu\nu}^{\text{field}}$: **Established semiclassical QFT** (given a field content and state).
- $Q_{\mu\nu}^{\text{spec}}$: **Speculative; must be quarantined.**

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3. Cosmology: Q^{geom}_{μν} in FLRW

We implemented the first concrete, executable piece: **Q^{geom}_{μν} in flat FLRW**.

3.1 Setup

Metric:

$$ds^2 = -dt^2 + a(t)^2 \delta_{ij} dx^i dx^j.$$

Hubble parameter:

$$H(t) = \frac{\dot{a}}{a}.$$

We focus on $f(R) = R + \alpha R^2$ with

$$\alpha = \frac{1}{12 m_0^2} \quad (M_{\mathrm{Pl}} = 1).$$

3.2 Explicit FLRW formulas for Q^{geom}

For the R+R² model in FLRW (C² term vanishes):

- **00-component:**

$$Q_{00}^{\mathrm{geom}} = 18\alpha \Big(6H^2 \dot{H} - \dot{H}^2 + 2H\ddot{H} \Big).$$

- **Spatial components:**

$$Q_{ij}^{\mathrm{geom}} = a^2 \delta_{ij}\, Q_{\mathrm{spatial}},$$
$$Q_{\mathrm{spatial}} = 6\alpha \Big(2\dddot{H} + 12H\ddot{H} + 9\dot{H}^2 + 18H^2\dot{H} \Big).$$

These formulas define the geometric quantum corrections in cosmology.

3.3 Modified Friedmann equation

With these corrections, the modified Friedmann equation (as implemented) is:

$$3H^2 + \Lambda = 8\pi G\, \rho_m + Q_{00}^{\mathrm{geom}} + Q_{00}^{\mathrm{field}}.$$

In code, we currently focus on the geometric piece and set $Q_{00}^{\mathrm{field}} = 0$ by default.

We expose a **diagnostic** function:

$$\text{residual}(H, \dot{H}, \ddot{H}; \rho_m, \Lambda) = 3H^2 + \Lambda - 8\pi G\rho_m - Q_{00}^{\text{geom}}.$$

Interpretation: - If residual = 0, the given (H, ρ_m, Λ) satisfy the modified Friedmann equation. - For a **GR-consistent** background without Q (i.e. $3H^2 + \Lambda = 8\pi G\rho_m$), the residual reduces to $-Q_{00}$, so its magnitude directly measures how much Q shifts the dynamics.

We explicitly **do not** claim to solve for $H(t)$ yet; we only evaluate Q on given backgrounds.

3.4 The CosmologicalQ Python class (concept)

Core features (implemented in `cosmology_q_calculator.py`):

- Constructor:
- Inputs: m_0, m_2 (in Planck units).
- Sets $\alpha = 1/(12m_0^2)$, $\beta = -1/(4m_2^2)$.
- Methods:
- `Q_geom_00(H, Hdot, Hddot)` – implements the exact formula above.
- `friedmann_residual(H, Hdot, Hddot, rho_m, Lambda)` – returns $3H^2 + \Lambda - 8\pi G\rho_m - Q_{00}$.
- `evaluate_on_toy_background(t_max, power)` – evaluates Q_{00} on a simple power-law background $a(t) = t^p$. **Explicit warning:** this does not solve the modified equations.
- `plot_toy_background(...)` – visualizes Q on this toy background with a disclaimer.
- `starobinsky_inflation_predictions(N)` – returns the standard (n_s, r) pair for Starobinsky inflation as a hard-coded benchmark (currently not derived from m_0 inside the code; clearly labeled as such).

3.5 Scaling results from the calculator

With realistic parameters:

- Starobinsky-like inflation: $m_0 \sim 1.3 \times 10^{-6} M_{\text{Pl}}$, $H_{\text{inf}} \sim 10^{-5} M_{\text{Pl}}$, $\epsilon \sim 10^{-3}$:
- Numerically, $Q_{00}^{\text{geom}}/(3H^2) \sim -0.18$.

- Interpretation: the R² term is an $O(1)$ contribution — it **drives** inflation, not a small correction.

- Late-time universe: $H_0 \sim 2.2 \times 10^{-61} M_{\text{Pl}}$:

- With the same m_0, we find

$$\frac{|Q_{00}^{\text{geom}}(t_0)|}{3H_0^2} \sim 10^{-110}.$$

- Interpretation: R² corrections are utterly negligible at late times.

This matches the expected EFT picture: - At high curvature (inflation scale), higher-curvature terms can dominate. - At extremely low curvature (today), they effectively turn off.

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4. Black Holes: Curvature Scaling Barrier & EFT-Only Pipeline

4.1 Curvature scaling barrier for Planck-triggered effects

For a Schwarzschild black hole of mass M in Planck units:

- Horizon radius: $r_s = 2M$.
- Kretschmann scalar:

$$K(r) = \frac{48M^2}{r^6}, \quad K_s \equiv K(r_s) = \frac{3}{4M^4}.$$

- Thus the ratio to Planck curvature ($K_{\rm Pl} \sim 1$) is

$$\frac{K_s}{K_{\rm Pl}} \sim \left(\frac{M_{\rm Pl}}{M}\right)^4.$$

For astrophysical black holes ($M \gg M_{\rm Pl}$), this is insanely small: - Stellar BH ($M \sim 10^{38} M_{\rm Pl}$): $K_s/K_{\rm Pl} \sim 10^{-152}$. - Supermassive BH ($M \sim 10^{47} M_{\rm Pl}$): $K_s/K_{\rm Pl} \sim 10^{-188}$.

Key conclusion: any modification that “turns on” at $K \sim K_{\rm Pl}$ is suppressed by $(M_{\rm Pl}/M)^4$ in BH exteriors and is **unobservable** for astrophysical BHs.

4.2 Death certificate for prime-weighted gravity (PWG) in BHs

Prime-weighted curvature models (PWG) introduced a factor

$$F(K) = \sum_{p \le P(K)} p^{-2}, \quad P(K) = \left\lfloor \frac{1}{\sqrt{K}} \right\rfloor.$$

In BH exteriors with $K \ll 1$, $P(K) \gg 1$ and the sum rapidly saturates:

$$F(K) \to F_0 \approx \zeta_{\rm primes}(2) \approx 0.45.$$

So for astrophysical BHs: - PWG reduces to a constant factor times a K term. - The **only** relevant scaling is $K \sim 1/M^4$. - The relative metric corrections at the photon sphere scale as

$$\frac{\delta A}{A} \sim \lambda K \sim \mathcal{O}(1) \times \frac{1}{M^4}.$$

This yields: - $\Delta R_{\rm shadow}/R_{\rm shadow} \sim (M_{\rm Pl}/M)^4$ — $\sim 10^{-152}$ for stellar BHs, $\sim 10^{-188}$ for SMBHs. - $\Delta\omega/\omega$ for QNMs with the same suppression.

Given EHT and LVK sensitivities (~10% at best), PWG-induced deviations are smaller by **150+ orders of magnitude**.

Verdict: - PWG in a Planck-triggered implementation is **phenomenologically inert** for BHs. - Any attempt to make it observable requires introducing new, fine-tuned curvature scales $K_* \ll K_{\rm Pl}$, which destroys the original motivation and demotes PWG to a generic, unprincipled deformation. - For BH phenomenology, PWG is declared **dead** and archived as a Planck-scale / early-universe curiosity only.

4.3 EFT-only BH pipeline

For BHs, we stick to the EFT action with (m_0, m_2) and build an observable pipeline:

1. **Metric ansatz**:

$$ds^2 = -A(r)dt^2 + B(r)^{-1}dr^2 + r^2 d\Omega^2,$$

$$A(r) = A_{\rm GR}(r)\Big[1 + \epsilon_0 a_0(x) + \epsilon_2 a_2(x)\Big],$$

$$B(r) = A_{\rm GR}(r)\Big[1 + \epsilon_0 b_0(x) + \epsilon_2 b_2(x)\Big],$$

   where:
2. $A_{\rm GR} = 1 - 2M/r$, $x = r/r_s$, $r_s = 2M$.
3. $\epsilon_i = 1/(m_i r_s)^2$.
4. a_0, a_2, b_0, b_2 are dimensionless deformation profiles from solving quadratic gravity's field equations.
5. **Photon sphere and shadow**:
6. Photon sphere $r_{\rm ph}$ from $A'(r)/A(r) = 2/r$.
7. Shadow radius $R_{\rm sh} = r_{\rm ph}/\sqrt{A(r_{\rm ph})}$.
8. Linearized deviations: $\Delta R_{\rm sh}/R_{\rm sh}^{(0)} = \kappa_0(M)\epsilon_0 + \kappa_2(M)\epsilon_2$.
9. **QNMs (eikonal)**:
10. Orbital frequency: $\Omega_{\rm ph} = \sqrt{A(r_{\rm ph})/r_{\rm ph}^2}$.
11. Lyapunov exponent: λ_L from second derivative of A/r^2 in tortoise coordinate.
12. Eikonal QNMs: $\omega_{\ell n} \approx \ell\Omega_{\rm ph} - i(n + 1/2)\lambda_L$.
13. Deviations again linear in $\epsilon_{0,2}$.
14. **Constraints**:
15. Map EHT and LVK deviations (consistent with 0 at ~10% level) to bounds on ϵ_0, ϵ_2 and thus m_0, m_2.
16. Currently: cosmology + binary pulsars give tighter bounds than BHs, so BHs are primarily cross-checks and future targets, not dominant constraints.

The key unification point: **the same m_0, m_2 used in cosmology** are used in BHs. This ties inflationary physics and BH corrections into a single EFT parameter set.

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5. Speculative Archive Rules

We define an explicit **Speculative Archive** with the following policies:

5.1 Prime-Weighted Gravity (PWG)

- Status: **Retired for BH phenomenology**.
- Reason: curvature scaling yields $\sim (M_{\text{Pl}}/M)^4$ suppression; unobservable for astrophysical BHs.
- Allowed domain: at most early-universe / Planck-scale toy models.
- Usage: may be mentioned as historical development or mathematical curiosity, never as a core prediction engine.

5.2 Novel entropy functionals

- Form: e.g. BH entropy$= A/4G\hbar + \int(\rho\ln\rho + I(\rho))$.
- Risk: can easily violate the **Generalized Second Law (GSL)** and first law if $I(\rho)$ is not derived from a consistent microstate counting.
- Status: **thermodynamically UNPROVEN**.
- Requirement before use:
- Show explicitly that proposed $I(\rho)$ respects GSL in basic processes (Hawking evaporation, matter infall, etc.).
- Until then, it cannot be treated as physically meaningful.

5.3 General policy

Speculative constructs: - Do **not** enter the core $Q^{\text{geom}}_{\mu\nu}$ or $Q^{\text{field}}_{\mu\nu}$ equations. - Must be clearly flagged as **Q^{\text{spec}}_{\mu\nu}** with explicit status. - Must not be used to claim observational signatures unless a full consistency and scaling analysis is done.

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6. Implementation Roadmap (Concrete)

Phase 1 – Framework template

- Create `Q_framework.tex` that:
- Defines the master action.
- Defines the $Q_{\mu\nu}$ decomposition.
- Lists parameter definitions and epistemic status.
- Include this section verbatim at the start of all new documents.

Phase 2 – Cosmology first (done in code, next in papers)

- `cosmology_q_calculator.py` implemented with:
- `CosmologicalQ(m0, m2)` class.
- Exact Q^{geom}_{00} and diagnostic methods.
- Write `Q_cosmology.tex` that:
- Presents the FLRW formulas.

- Shows inflation vs late-time scaling.
- Documents the calculator usage and outputs.

Phase 3 – BH consistency check

- Implement BH EFT metric deformation using same m_0, m_2.
- Build Python tools (analogous to the cosmology calculator) to:
- Take metric functions $A(r), B(r)$ as input.
- Compute photon sphere, shadow radius, and eikonal QNMs.
- Show explicitly that for realistic m_0, m_2 consistent with cosmology, BH deviations are $\ll 10\%$ and consistent with current null results.

Phase 4 – Refactor existing documents

- Update Starobinsky, ToE, wormhole, and TEGR documents to:
- Reference the canonical Q-framework.
- Use the $Q^{\rm geom}/Q^{\rm field}/Q^{\rm spec}$ split.
- Drop or quarantine speculative sectors (e.g., PWG in BHs) according to the archive rules.

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7. What This Achieves

- A **coherent, reusable $Q_{\mu\nu}$ framework** anchored in EFT and semiclassical gravity.
- A **working cosmology calculator** that:
- Shows higher-curvature terms drive inflation but die away at late times.
- Is ready to be extended and reused.
- A **clear verdict on PWG** and similar numerological constructs:
- BH phenomenology: dead by curvature scaling.
- Allowed only as quarantined speculation in regimes where we genuinely lack better tools.
- A **methodological foundation** where every new extension must:
- Be placed into the $Q^{\rm geom}/Q^{\rm field}/Q^{\rm spec}$ taxonomy.
- Pass scaling and consistency checks before being used for predictions.

This is the baseline spec going forward: no new work should bypass or contradict this structure without an explicit, justified revision.